

Problem Statement

You are working as a Data Engineer for a large scale food delivery platform similar to Swiggy. The platform operates across multiple cities and manages customers, restaurants, menu items, delivery partners, orders, and payments. The company wants a robust relational database system that supports daily transactions, ensures data integrity, and enables business reporting.

Your task is to design and operate the database using DDL and DML.

Business Requirements

1. Customers can register with name, email, phone number, and address.
2. Each customer must have a unique email and phone number.
3. Restaurants operate in specific cities and have a rating between 0 and 5.
4. Restaurants can be active or inactive. Only active restaurants can receive orders.
5. Each restaurant can have multiple menu items.
6. Menu items must store name, price, and availability status.
7. Delivery partners can register with vehicle type and availability status.
8. Orders are placed by customers from a single restaurant.
9. An order can contain multiple menu items with quantities.
10. Each order must have a delivery partner assigned.
11. Payments must be recorded for every order with payment method and status.

Part A DDL Tasks

Design the complete schema for the platform.

You must

- Create all required tables
- Define primary keys

- Define foreign keys
- Apply constraints such as unique, not null, check
- Enforce referential integrity
- Ensure business rules are validated at database level

Schema Hint

customers

customer_id primary key
name
email unique
phone unique
address
created_at

restaurants

restaurant_id primary key
name
city
rating check between 0 and 5
is_active

menu_items

item_id primary key
restaurant_id foreign key references restaurants
item_name
price
is_available

delivery_partners

partner_id primary key
name
phone unique
vehicle_type
is_available

orders

order_id primary key
customer_id foreign key references customers
restaurant_id foreign key references restaurants
partner_id foreign key references delivery_partners

order_time
total_amount
order_status

order_items
order_item_id primary key
order_id foreign key references orders
item_id foreign key references menu_items
quantity

payments
payment_id primary key
order_id foreign key references orders
payment_method
payment_status
payment_time